

Отчет по лабораторной работе №3

Настройка DHCP-сервера

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Цель работы

Приобретение практических навыков по установке и конфигурированию DHCP-сервера.

Задание

1. Установить на виртуальной машине server DHCP-сервер.
2. Настроить виртуальную машину server в качестве DHCP-сервера для виртуальной внутренней сети.
3. Получить параметры от DHCP-сервера в виртуальной внутренней сети путём запуска виртуальной машины client и применения соответствующих утилит диагностики.
4. Настроить обновление DNS-зоны при появлении в виртуальной внутренней сети новых узлов.
5. Проверить корректность работы DHCP-сервера и обновления DNS-зоны в виртуальной внутренней сети путём запуска виртуальной машины client и применения соответствующих утилит диагностики.
6. Написать скрипт для Vagrant, фиксирующий действия по установке и настройке DHCP-сервера во внутреннем окружении виртуальной машины server. Внести соответствующие изменения в Vagrantfile.

Выполнение лабораторной работы

Подключение к виртуальной
машине server

Запускаем виртуальную машину server и подключаемся к ней по SSH.

```
D:\work\mavinogradova>cd vagrant
D:\work\mavinogradova\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
==> server: Machine booted and ready!
==> server: Checking for guest additions in VM...
==> server: Setting hostname...
```

Рис. 1: Запуск виртуальной машины server

Установка DHCP-сервера

Входим под пользователем mavinogradova и переходим в режим суперпользователя.

```
D:\work\mavinogradova\vagrant>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 12:01:06 2026 from 10.0.2.2
vagrant@server:~$ su- mavinogradova
bash: su-: command not found...
^[[A^[[B^Z
[1]+  Stopped                  su- mavinogradova
vagrant@server:~$ su -mavinogradova
su: invalid option -- 'a'
Try 'su --help' for more information.
vagrant@server:~$ su - mavinogradova
Password:
Last login: Sat Sep 12 10:40:42 UTC 2026 on pts/0
[mavinogradova@server.mavinogradova.net ~]$ sudo-i
bash: sudo-i: command not found...
[mavinogradova@server.mavinogradova.net ~]$ sudo -i
[sudo] password for mavinogradova:
Sorry, try again.
[sudo] password for mavinogradova:
```

Рис. 2: Вход под mavinogradova и получение root

Устанавливаем DHCP-сервер Kea.

```
[root@server.mavinogradova.net ~]# dnf-y install kea
bash: dnf-y: command not found...
[root@server.mavinogradova.net ~]# dnf -y install kea
Extra Packages for Enterprise Linux 10 - x86_64          9.2 kB/s | 30 kB   00:03
Extra Packages for Enterprise Linux 10 - x86_64          1.7 MB/s | 6.7 MB   00:03
Rocky Linux 10 - BaseOS                                191 B/s | 4.3 kB   00:23
Rocky Linux 10 - BaseOS                                890 kB/s | 28 MB   00:32
Rocky Linux 10 - AppStream                             2.8 kB/s | 4.4 kB   00:01
Rocky Linux 10 - AppStream                             2.0 MB/s | 2.7 MB   00:01
Rocky Linux 10 - CRB                                    11 kB/s | 4.3 kB   00:00
Rocky Linux 10 - CRB                                    3.0 kB/s | 2.0 MB   00:00
```

Рис. 3: Установка kea

Конфигурирование DHCP-сервера

Сохраняем резервную копию конфигурационного файла и открываем его на редактирование.

```
[root@server.mavinoegradova.net ~]# cp /etc/kea/kea-dhcp4.conf /etc/kea/kea-dhcp4.conf_$(date -I)
[root@server.mavinoegradova.net ~]# nano /etc/kea/kea-dhcp4.conf
```

Рис. 4: Резервная копия kea-dhcp4.conf

Анализируем содержимое /etc/kea/kea-dhcp4.conf.

```
GNU nano 8.1 /etc/kea/kea-dhcp4.conf
{
"Dhcp4": {
  "interfaces-config": {
    "interfaces": [ "eth1" ]
  },
  "control-socket": {
    "socket-type": "unix",
    "socket-name": "kea4-ctrl-socket"
  },
  "lease-database": {
    "type": "memfile",
    "lfc-interval": 3600
  },
  "expired-leases-processing": {
    "reclaim-timer-wait-time": 10,
    "flush-reclaimed-timer-wait-time": 25,
    "hold-reclaimed-time": 3600,
    "max-reclaim-leases": 100,
    "max-reclaim-time": 250,
    "unwarned-reclaim-cycles": 5
  },
  "renew-timer": 900,
```

Рис. 5: Содержимое kea-dhcp4.conf

Задаём конфигурацию DHCP-сети и проверяем конфигурационный файл.

```
[root@server.mavinoogradova.net ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-19 12:56:04.912 WARN [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_RT_DISABLED_QUEUE_CONTROL disabling dhcp
queue control when multi-threading is enabled.
2026-09-19 12:56:04.913 WARN [kea-dhcp4.dhcp4/13563.140174058039424] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-thre
ading is enabled and host reservations lookup is always performed first.
2026-09-19 12:56:04.913 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been
added to configuration: 192.168.1.0/24 with params: t1=900, t2=1800, valid-lifetime=3600
2026-09-19 12:56:04.915 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_CFGMGR_SOCKET_TYPE_SELECT using socket t
ype raw
2026-09-19 12:56:04.923 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface
eth1
2026-09-19 12:56:04.924 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-
type" not specified, using default socket type raw
2026-09-19 12:56:04.924 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the follow
ing lease backend types are available: memfile
2026-09-19 12:56:04.924 INFO [kea-dhcp4.dhcp4/13563.140174058039424] HOSTS_BACKENDS_REGISTERED the following host backe
nd types are available:
2026-09-19 12:56:04.925 INFO [kea-dhcp4.dhcp4srv/13563.140174058039424] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the follow
ing forensic backend types are available:
2026-09-19 12:56:04.926 INFO [kea-dhcp4.dhcp4/database/13563.140174058039424] CONFIG_BACKENDS_REGISTERED the following config
backend types are available:
[root@server.mavinoogradova.net ~]#
```

Рис. 6: Проверка конфигурационного файла

Перезагружаем конфигурацию dhcpd и включаем автозапуск.

```
[root@server.mavinogradova.net ~]# systemctl --system daemon-reload
[root@server.mavinogradova.net ~]# systemctl enable kea-dhcpd.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcpd.service' + '/usr/lib/systemd/system/kea-dhcpd.service'.
[root@server.mavinogradova.net ~]#
```

Рис. 7: Перезагрузка конфигурации и включение автозапуска

Добавляем запись для DHCP-сервера в DNS-зоны (прямую и обратную).

```
GNU nano 8.1 /var/named/master/fz/mavinogradova.net Modified
$TTL 1D
@ IN SOA @ server.mavinogradova.net. (
    2026091913 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ) ; minimum
NS @
A 192.168.1.1

$ORIGIN mavinogradova.net.
server A 192.168.1.1
ns A 192.168.1.1
dhcp A 192.168.1.1
```

Рис. 8: Прямая DNS-зона mavinogradova.net

```
GNU nano 8.1 /var/named/master/rz/192.168.1 Modified
$TTL 1D
@ IN SOA @ server.mavinogradova.net. (
    2026091913 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ) ; minimum
NS @
A 192.168.1.1

$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.mavinogradova.net.
1 PTR ns.mavinogradova.net.
1 PTR dhcp.mavinogradova.net.
```

Рис. 9: Обратная DNS-зона 192.168.1

Перезапускаем named и проверяем ping dhcp.mavinogradova.net.

```
[root@server.mavinogradova.net ~]# systemctl restart named
[root@server.mavinogradova.net ~]# ping dhcp.user.net
ping: dhcp.user.net: Name or service not known
[root@server.mavinogradova.net ~]# ping dhcp.mavinogradova.net
PING dhcp.mavinogradova.net (192.168.1.1) 56(84) bytes of data.
64 bytes from ns.mavinogradova.net (192.168.1.1): icmp_seq=1 ttl=64 time=3.99 ms
64 bytes from ns.mavinogradova.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.193 ms
64 bytes from ns.mavinogradova.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.476 ms
64 bytes from ns.mavinogradova.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.099 ms
^C
--- dhcp.mavinogradova.net ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3047ms
rtt min/avg/max/mdev = 0.099/1.189/3.991/1.623 ms
[root@server.mavinogradova.net ~]#
```

Рис. 10: Перезапуск named и проверка ping dhcp.mavinogradova.net

Разрешаем работу с DHCP в межсетевом экране.

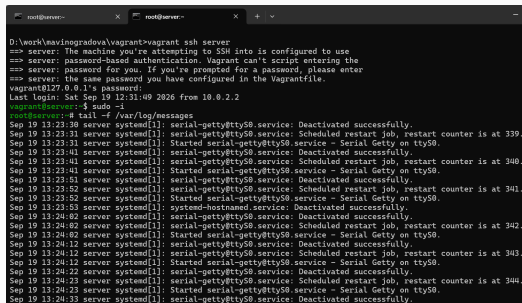
```
[root@server.navigradska.net ~]# firewall-cmd --list-services
cockpit dhcpd-client dnsmasq
[root@server.navigradska.net ~]# firewall-cmd --add-service=cockpit
success
[root@server.navigradska.net ~]# firewall-cmd --add-service=cockpit --permanent
success
[root@server.navigradska.net ~]# firewall-cmd --add-service=cockpit --permanent
success
[root@server.navigradska.net ~]#
```

Рис. 11: Разрешение DHCP в firewall

```
[root@server.mavinogradova.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.mavinogradova.net ~]# restorecon -vR /var/named
[root@server.mavinogradova.net ~]# restorecon -vR /var/lib/kea/
[root@server.mavinogradova.net ~]#
```

Рис. 12: Восстановление контекста SELinux

Запускаем мониторинг системных сообщений.



```
root@server:~  
D:\work\mavinoogradova\vagrant>vagrant ssh server  
==> server: The machine you're attempting to SSH into is configured to use  
==> server: password-based authentication. Vagrant can't script entering the  
==> server: password for you. If you're prompted for a password, please enter  
==> server: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sat Sep 19 12:31:49 2026 from 10.0.2.2  
vagrant@server:~$ sudo -i  
root@server:~# tail -f /var/log/messages  
Sep 19 13:23:30 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:23:31 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 339.  
Sep 19 13:23:31 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:23:41 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:23:41 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 340.  
Sep 19 13:23:41 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:23:51 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:23:52 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 341.  
Sep 19 13:23:52 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:23:53 server systemd[1]: systemd-hostnamed.service: Deactivated successfully.  
Sep 19 13:24:02 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:24:02 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 342.  
Sep 19 13:24:02 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:24:12 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:24:12 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 343.  
Sep 19 13:24:12 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:24:22 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.  
Sep 19 13:24:23 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 344.  
Sep 19 13:24:23 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.  
Sep 19 13:24:33 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
```

Рис. 13: Мониторинг системных сообщений

Запускаем DHCP-сервер и убеждаемся в успешном запуске.

```
[root@server.mavinogradova.net ~]# systemctl start kea-dhcp4.service  
[root@server.mavinogradova.net ~]# |
```

Рис. 14: Запуск DHCP-сервера

```
RTING Kea DHCPv4 server version 3.0.3 (stable) starting  
Sep 19 13:24:35 server kea-dhcp4[14386]: 2026-09-19 13:24:35.226 INFO [kea-dhcp4.commands/14386.140580287117uu0] COMMAND_RECEIVED Received command 'config-set'  
Sep 19 13:24:35 server systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
```

Рис. 15: Запуск kea-dhcp4.service

Анализ работы DHCP-сервера

Создаём файл 01-routing.sh в каталоге provision/client.

```
D:\work\mavinogradova\vagrant\provision\client>type nul > 01-routing.sh

D:\work\mavinogradova\vagrant\provision\client>dir
Том в устройстве D имеет метку Data
Серийный номер тома: C21C-2008

Содержимое папки D:\work\mavinogradova\vagrant\provision\client

10.09.2026  19:16    <DIR>          .
10.09.2026  19:11    <DIR>          ..
10.09.2026  17:13                44 01-dummy.sh
19.09.2026  16:34                0 01-routing.sh
                2 файлов                44 байт
                2 папок 115 302 137 856 байт свободно

D:\work\mavinogradova\vagrant\provision\client>
```

Рис. 16: Создание файла 01-routing.sh

Прописываем в нём скрипт, изменяющий настройки NetworkManager.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\virt> cd D:\work\mavinogradova\vagrant\provision\client
PS D:\work\mavinogradova\vagrant\provision\client> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

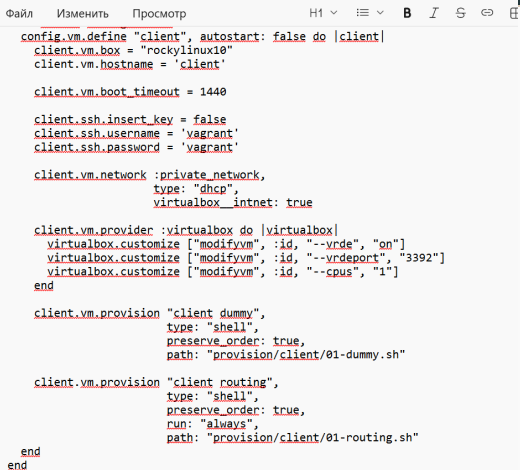
PS D:\work\mavinogradova\vagrant\provision\client> pwd

Path
----
D:\work\mavinogradova\vagrant\provision\client

PS D:\work\mavinogradova\vagrant\provision\client> $content = @'
>> #!/bin/bash
>> echo "Provisioning script $0"
>> nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
>> nmcli connection up "eth1"
>> nmcli connection modify eth0 ipv4.never-default true
>> nmcli connection modify eth0 ipv6.never-default true
>> nmcli connection down eth0
>> nmcli connection up eth0
>> '@
PS D:\work\mavinogradova\vagrant\provision\client>
PS D:\work\mavinogradova\vagrant\provision\client> $content = $content -replace "`n", "`r`n"
PS D:\work\mavinogradova\vagrant\provision\client> [System.IO.File]::WriteAllText("$PWD\01-routing.sh", $content, [System.Text.UTF8Encoding]::new($false))
PS D:\work\mavinogradova\vagrant\provision\client> dir 01-routing.sh
```

Рис. 17: Содержимое 01-routing.sh

Подключаем скрипт в Vagrantfile в разделе конфигурации для клиента.



```
config.vm.define "client", autostart: false do |client|
  client.vm.box = "rockylinux10"
  client.vm.hostname = 'client'

  client.vm.boot_timeout = 1440

  client.ssh.insert_key = false
  client.ssh.username = 'vagrant'
  client.ssh.password = 'vagrant'

  client.vm.network :private_network,
    type: "dhcp",
    virtualbox____intnet: true

  client.vm.provider :virtualbox do |virtualbox|
    virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
    virtualbox.customize ["modifyvm", :id, "--vrdeport", "3392"]
    virtualbox.customize ["modifyvm", :id, "--cpus", "1"]
  end

  client.vm.provision "client_dummy",
    type: "shell",
    preserve_order: true,
    path: "provision/client/01-dummy.sh"

  client.vm.provision "client_routing",
    type: "shell",
    preserve_order: true,
    run: "always",
    path: "provision/client/01-routing.sh"
end
end
```

Рис. 18: Фрагмент Vagrantfile с client routing

Запускаем клиента с провижинингом.

```
D:\work\mavinogradova\vagrant>vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Clearing any previously set forwarded ports...
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
      client: Adapter 1: nat
      client: Adapter 2: intnet
==> client: Forwarding ports...
      client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
      client: SSH address: 127.0.0.1:2200
      client: SSH username: vagrant
      client: SSH auth method: password
==> client: Machine booted and ready!
==> client: Checking for guest additions in VM...
      client: Guest additions not detected. Not required for this VM. Guest
```

Рис. 19: Запуск клиента с провижинингом

Наблюдаем логи DHCP-сервера о выдаче IP-адреса.

```
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the following lease backend types are available: memfile
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.hosts.140638289537152] HOSTS_BACKENDS_REGISTERED the following host backend types are available:
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the following forensic backend types are available:
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.database.140638289537152] CONFIG_BACKENDS_REGISTERED the following config backend types are available:
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4.140638289537152] DHCP4_CONFIG_COMPLETE DHCPv4 server has completed configuration: added IPv4 subnets: 1; DDNS: disabled
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_MEMFILE_DB opening memory file lease database: lfc-interval=3600 typememfile universe=4
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_MEMFILE_LEASE_FILE_LOAD loading leases from file /var/lib/kea/kea-leases4.csv.2
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_MEMFILE_LEASE_FILE_LOAD loading leases from file /var/lib/kea/kea-leases4.csv
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_MEMFILE_EXTRACT_EXTENDED_INFO extracting extended info saw 1 leases, extended info sanity checks modified 0 / updated 0 leases and 0 leases have relay or remote id
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_MEMFILE_LFC_SETUP setting up the Lease File Cleanup interval to 3600 sec
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4srv.140638289537152] DHCP4_SRV_CFGMGR_USE_ALLOCATOR using the iterative allocator for V4 leases in subnet 192.168.1.0/24
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4.140638289537152] DHCP4_MULTI_THREADING_INFO enabled: yes, number of threads: 2, queue size: 64
Sep 19 15:05:42 server kea-dhcp4[15677]: INFO [kea-dhcp4.dhcp4.140638289537152] DHCP4_STARTED Kea DHCPv4 server version 3.0.3 started
Sep 19 15:05:42 server systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
Sep 19 15:05:58 server systemd[1]: serial-getty@tty50.service: Deactivated successfully.
Sep 19 15:05:51 server systemd[1]: serial-getty@tty50.service: Scheduled restart job, restart counter is at 929.
```

Рис. 20: Логи DHCP-сервера

Информация о работе DHCP-сервера в файле /var/lib/kea/kea-leases4.csv.

```
[root@server.mavinogradova.net ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
192.168.1.30,08:00:27:23:bf:22,01:08:00:27:23:bf:22,3600,1789831476,1,0,0,client.mavinogradova.net,0,,0
192.168.1.30,08:00:27:23:bf:22,01:08:00:27:23:bf:22,3600,1789832783,1,0,0,client.mavinogradova.net,0,,0
192.168.1.30,08:00:27:23:bf:22,01:08:00:27:23:bf:22,3600,1789833951,1,0,0,client.mavinogradova.net,0,,0
[root@server.mavinogradova.net ~]#
```

Рис. 21: Содержимое kea-leases4.csv

Входим на client и вводим ifconfig.

```
vagrant@client:~$ su - mavinogradova
Password:
Last login: Fri Sep 11 17:15:56 UTC 2026 on pts/0
mavinogradova@client:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:628c:f837:2:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x0<global>
    inet6 fe80:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:64:52:85 txqueuelen 1000 (Ethernet)
    RX packets 2399 bytes 228308 (222.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1646 bytes 222476 (217.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.30 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80:bb29:a83f:3d1f:e5b prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:23:bf:22 txqueuelen 1000 (Ethernet)
    RX packets 57 bytes 8081 (7.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 405 bytes 36158 (35.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 18 bytes 2118 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 2118 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mavinogradova@client:~$ |
```

Рис. 22: Вывод ifconfig на клиенте

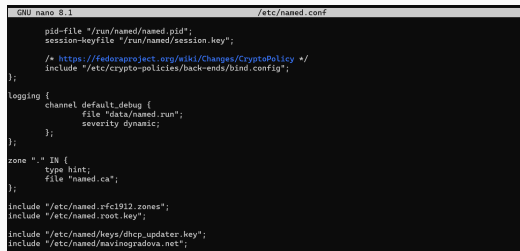
Настройка обновления DNS-зоны

Создаём TSIG-ключ на сервере с Bind9.

```
[root@server.mavinogradova.net ~]# mkdir -p /etc/named/keys
[root@server.mavinogradova.net ~]# tsig-keygen -a HMAC-SHA512 DHCP_UPDATE > /etc/named/keys/dhcp_updater.key
[root@server.mavinogradova.net ~]# cat /etc/named/keys/dhcp_updater.key
key "DHCP_UPDATE" {
    algorithm hmac-sha512;
    secret "bIo+P2hFohceUw97weIdnZwYooUMZeSHI5xDxxkQ1JTukM/qBFcSRHMUSkPIHbu0QJcP/BxBR2i+U0pTt8iemw==";
};
[root@server.mavinogradova.net ~]# chown -R named:named /etc/named/keys
[root@server.mavinogradova.net ~]# include "/etc/named/keys/dhcp_updater.key";
bash: include: command not found...
[root@server.mavinogradova.net ~]# ^C
[root@server.mavinogradova.net ~]# nano /etc/named.conf
```

Рис. 23: Генерация TSIG-ключа

Поправляем права доступа на ключ.



```
GNU nano 8.1 /etc/named.conf

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default.debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

include "/etc/named/keys/dhcp_updater.key";
include "/etc/named/maivogradova.net";
```

Рис. 24: Права доступа на ключ

Подключаем ключ в файле /etc/named.conf.

```
zone "mavinogradova.net" IN {
    type primary;
    file "master/fz/mavinogradova.net";
    update-policy {
        grant DHCP_UPDATE wildcard *.mavinogradova.net A DHCID;
    };
};

zone "1.168.192.in-addr.arpa" IN {
    type primary;
    file "master/rz/192.168.1";
    update-policy {
        grant DHCP_UPDATE wildcard *.1.168.192.in-addr.arpa PTR DHCID;
    };
};
```

Рис. 25: Подключение ключа в named.conf

Разрешаем обновление зон в /etc/named/mavinogradova.net.

```
[root@server.mavinogradova.net ~]# named-checkconf
[root@server.mavinogradova.net ~]# systemctl restart named
[root@server.mavinogradova.net ~]# systemctl status named --no-pager
● named.service - Berkeley Internet Name Domain (DNS)
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-19 15:39:16 UTC; 23s ago
```

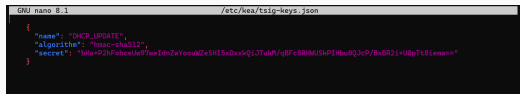
Рис. 26: Разрешение обновления зон

Проверяем конфигурационный файл и перезапускаем DNS-сервер.

```
[root@server.mavinogradova.net ~]# touch /etc/kea/tsig-keys.json
[root@server.mavinogradova.net ~]# nano /etc/kea/tsig-keys.json
[root@server.mavinogradova.net ~]# cat /etc/named/keys/dhchap_updater.key
key "DHCP_UPDATE" {
    algorithm hmac-sha512;
    secret "bko+P2hf0hceUw97weIdnZwYooUMZeSHI5xDxxkQ1JTukM/qBFc5RHWUSkPIHbu0QJcP/BxBR2i+U0pTt8iemw==";
};
[root@server.mavinogradova.net ~]# nano /etc/kea/tsig-keys.json
```

Рис. 27: Проверка named.conf и перезапуск named

Формируем ключ для Kea — /etc/kea/tsig-keys.json.



```
GNU nano 8.1 /etc/kea/tsig-keys.json
{
  "name": "DNS_UPDATE",
  "algorithm": "hmac-sha512",
  "secret": "bMo+P2hFahceUk97weIdnZwYoonWZeSHISxDxxkQlJTukM/qBFc5RHMUSkPIHbu0QJcP/BxBR2i+U8pTt8Iem=="
}
```

Рис. 28: Создание tsig-keys.json

Переносим ключ и переписываем его в формате JSON.

```
[root@server.mavinogradova.net ~]# chown kea:kea /etc/kea/tsig-keys.json  
[root@server.mavinogradova.net ~]# chmod 640 /etc/kea/tsig-keys.json  
[root@server.mavinogradova.net ~]#
```

Рис. 29: Содержимое tsig-keys.json

Сменим владельца и поправим права доступа.

```
// This is a basic configuration for the Kea DHCP DDNS daemon.
{
  "DhcpDdns":
  {
    "ip-address": "127.0.0.1",
    "port": 53001,
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "/run/kea/kea-ddns-ctrl-socket"
    },
    "tsig-keys": [ <?include "/etc/kea/tsig-keys.json"?> ],
    "forward-ddns": {
      "ddns-domains": [
        {
          "name": "mavinogradova.net.",
          "key-name": "DHCP_UPDATE",
          "dns-servers": [
            { "ip-address": "192.168.1.1" }
          ]
        }
      ]
    }
  }
},
```

Рис. 30: Права на tsig-keys.json

Настройка в файле /etc/kea/kea-dhcp-ddns.conf.

```
[root@server.mavinsgradova.net ~]# chmod kea-kea /etc/kea/kea-dhcp-ddns.conf
[root@server.mavinsgradova.net ~]# kea-dhcp-ddns-t /etc/kea/kea-dhcp-ddns.conf
bash: kea-dhcp-ddns-t: command not found
[root@server.mavinsgradova.net ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2026-09-19 16:04:47.218 INFO [kea-dhcp-ddns.dctl/16753:16077085202128] DCTL_CONFIG_CHECK_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), testConfiguration check successful
[root@server.mavinsgradova.net ~]# systemctl enable --now kea-dhcp-ddns.service
libsystemd command 'enable --now' did you mean 'enable'?
[root@server.mavinsgradova.net ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service → /usr/lib/systemd/system/kea-dhcp-ddns.service.
[root@server.mavinsgradova.net ~]# systemctl status kea-dhcp-ddns.service
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-19 16:05:38 UTC; 9s ago
     Invocation: 066e2186ab7b4e91aa8be118521edcdc
       Docs: man:kea-dhcp-ddns(8)
     Main PID: 16753 (kea-dhcp-ddns)
        Status: "Dispatching packets..."
         Tasks: 0 (limit: 10229)
       Memory: 2.3M (peak: 7M)
          CPU: 12ms
     CGroup: /system.slice/kea-dhcp-ddns.service
             └─16753 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 19 16:05:38 server.mavinsgradova.net systemd[1]: Starting kea-dhcp-ddns.service - Kea DHCP-DDNS Server...
Sep 19 16:05:38 server.mavinsgradova.net kea-dhcp-ddns[16753]: 2026-09-19 16:05:38.903 INFO [kea-dhcp-ddns.dctl/16753:16077085202128] DCTL_CONFIG_CHECK_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), testConfiguration check successful
Sep 19 16:05:38 server.mavinsgradova.net kea-dhcp-ddns[16753]: INFO COMMAND_ACCEPTOR_START Starting to accept connections
Sep 19 16:05:38 server.mavinsgradova.net kea-dhcp-ddns[16753]: INFO DCTL_CONFIG_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), testConfiguration check successful
Sep 19 16:05:38 server.mavinsgradova.net kea-dhcp-ddns[16753]: INFO DHCP_DNS_STARTED Kea DHCP-DDNS server version 3.0.0
Sep 19 16:05:38 server.mavinsgradova.net systemd[1]: Started kea-dhcp-ddns.service - Kea DHCP-DDNS Server.
```

Рис. 31: Настройка kea-dhcp-ddns.conf

Проверяем синтаксис, запускаем службу ddns и проверяем статус.

```
        {
            "name": "routers",
            "data": "192.168.1.1"
        }
    ],
    "dhcp-ddns": {
        "enable-updates": true
    },
    "ddns-qualifying-suffix": "mavinogradova.net",
    "ddns-override-client-update": true,
    "loggers": [
        {
            "name": "kea-dhcp4",
            "output-options": [
                {
                    "output": "syslog"
                }
            ],
            "severity": "INFO",
            "debugLevel": 0
        }
    ]
}
```

Рис. 32: Проверка и запуск kea-dhcp-ddns

Включаем enable-updates в /etc/kea/kea-dhcp4.conf.

```
*[[A[root@server.navinogradova.net ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-19 16:10:36.728 WARN [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_MT_DISABLED_QUEUE_CONTROL disabling dhcp
queue control when multi-threading is enabled.
2026-09-19 16:10:36.729 WARN [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-thre
ading is enabled and host reservations lookup is always performed first.
2026-09-19 16:10:36.729 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been
added to configuration: 192.168.1.0/24 with params: tl=900, t2=1800, valid-lifetime=3600
2026-09-19 16:10:36.730 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_CFGMGR_SOCKET_TYPE_SELECT using socket t
ype raw
2026-09-19 16:10:36.732 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface
eth1
2026-09-19 16:10:36.732 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-
type" not specified, using default socket type raw
2026-09-19 16:10:36.732 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the follow
ing lease backend types are available: newfile
2026-09-19 16:10:36.732 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_HOSTS_BACKENDS_REGISTERED the following host backe
nd types are available:
2026-09-19 16:10:36.732 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the followi
ng forensic backend types are available:
2026-09-19 16:10:36.733 INFO [kea-dhcp4.dhcp4/16853.139766320928896] DHCP4_SRV_CONFIG_BACKENDS_REGISTERED the following config
backend types are available:
[root@server.navinogradova.net ~]#
```

Рис. 33: Правка kea-dhcp4.conf (enable-updates)

Проверяем и перезапускаем DHCP-сервер.

```
[root@server.mavinogradova.net ~]# systemctl restart kea-dhcp4.service
[root@server.mavinogradova.net ~]# systemctl status kea-dhcp4.service
* kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-19 16:11:28 UTC; 5s ago
   Invocation: blaa557b55164856a642b8f38247796
   Docs: man:kea-dhcp4(8)
   Main PID: 16872 (kea-dhcp4)
   Status: "Dispatching packets..."
   Tasks: 7 (limit: 10259)
   Memory: 2.5M (peak: 6.9M)
   CPU: 166ms
   CGroup: /system.slice/kea-dhcp4.service
           └─16872 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_MEMFILE_DBG
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_MEMFILE_LE
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_MEMFILE_EX
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_MEMFILE_LF
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_DHCP_DONS
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcpdsv.139624472541312] DHCP4_SRV_CFGMGR_USE
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcp4.139624472541312] DHCP4_MULTI_THREADING
Sep 19 16:11:28 server.mavinogradova.net kea-dhcp4[16872]: INFO [kea-dhcp4.dhcp4.139624472541312] DHCP4_STARTED Kea DHCP
Sep 19 16:11:28 server.mavinogradova.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
```

Рис. 34: Проверка и перезапуск kea-dhcp4

На машине client переполучаем адрес.

```
root@client:~# nmcli connection down eth1
Connection 'eth1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
root@client:~# nmcli connection up eth1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
root@client:~#
```

Рис. 35: Переполучение адреса на клиенте

Появление файла mavinogradova.net.jnl в каталоге прямой зоны.

```
[root@server.mavinogradova.net ~]# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
192.168.1.30,08:00:27:23:bf:22,01:08:00:27:23:bf:22,3600,1789837851,1,0,0,client.mavinogradova.net,0,0
192.168.1.30,08:00:27:23:bf:22,01:08:00:27:23:bf:22,3600,1789838381,1,1,1,client.mavinogradova.net,0,0
[root@server.mavinogradova.net ~]# ls -la /var/named/master/fz/
total 8
drwxr-xr-x. 2 named named 60 Sep 19 16:19 .
drwxr-xr-x. 4 named named 26 Sep 12 11:33 ..
-rw-r-----. 1 named named 486 Sep 19 13:08 mavinogradova.net
-rw-r-----. 1 named named 843 Sep 19 16:19 mavinogradova.net.jnl
[root@server.mavinogradova.net ~]#
```

Рис. 36: Появление mavinogradova.net.jnl

Анализ работы DHCP-сервера после настройки DDNS

Убеждаемся в наличии DNS-записи о клиенте через dig.

```
root@client:~# dig @192.168.1.1 client.mavinogradova.net

; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.mavinogradova.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 52334
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d7dcd1f1eae0995010000006aaeb6ff00e0333f75b7fbd0 (good)
;; QUESTION SECTION:
;client.mavinogradova.net.      IN      A

;; ANSWER SECTION:
client.mavinogradova.net. 1200 IN      A      192.168.1.30

;; Query time: 6 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Sat Sep 19 16:23:27 UTC 2026
;; MSG SIZE rcvd: 97

root@client:~#
```

Рис. 37: Проверка DNS-записи клиента через dig

Внесение изменений в настройки внутреннего окружения

Копируем конфигурационные файлы DHCP в каталог provision.

```
[root@server.mavinogradova.net ~]# cd /vagrant/provision/server
[root@server.mavinogradova.net server]# mkdir -p /vagrant/provision/server/dhcp/etc/kea
bash: mkdir -p: command not found...
^[[B[root@server.mavinogradova.net server]# ^C
[root@server.mavinogradova.net server]# mkdir -p /vagrant/provision/server/dhcp/etc/kea
[root@server.mavinogradova.net server]# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/
```

Рис. 38: Копирование конфигурационных файлов DHCP

Копируем конфигурационные файлы DNS-сервера.

```
[root@server.navinogradova.net server]# cd /vagrant/provision/server/dns/
[root@server.navinogradova.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/navinogradova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/sz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error
[root@server.navinogradova.net dns]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
bash: cp-R: command not found...
[root@server.navinogradova.net dns]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
cp: overwrite '/vagrant/provision/server/dns/etc/named/navinogradova.net'? y
```

Рис. 39: Копирование конфигурационных файлов DNS

Создаём исполняемый файл dhcp.sh.

```
[root@server.mavinogradova.net dns]# cd /vagrant/provision/server  
[root@server.mavinogradova.net server]# touch dhcp.sh  
[root@server.mavinogradova.net server]# chmod +x dhcp.sh  
[root@server.mavinogradova.net server]#
```

Рис. 40: Создание dhcp.sh

Прописываем в нём скрипт для Vagrant.

```
#!/bin/bash
echo "Provisioning script \${0}"
echo "Install needed packages"
dnf -y install kea
echo "Copy configuration files"
cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/
echo "Fix permissions"
chown -R kea:kea /etc/kea
chmod 640 /etc/kea/tsig-keys.json
restorecon -vR /etc
restorecon -vR /var/lib/kea
echo "Configure firewall"
firewall-cmd --add-service dhcp
firewall-cmd --add-service dhcp --permanent
echo "Start dhcpd service"
systemctl --system daemon-reload
```

Рис. 41: Содержимое dhcp.sh

Добавляем строку `server.vm.provision "server dhcp"` в Vagrantfile.

```
        virtualbox__intnet: true

server.vm.provider :virtualbox do |virtualbox|
  virtualbox.customize ["modifyvm", :id, "--vrde", "on"]
  virtualbox.customize ["modifyvm", :id, "--vrdeport", "3391"]
end

server.vm.provision "server dummy",
  type: "shell",
  preserve_order: true,
  inline: "echo 'server dummy provisioner executed'"

server.vm.provision "server dns",
  type: "shell",
  preserve_order: true,
  inline: "echo 'server dns provisioner executed'"

server.vm.provision "server dhcp",
  type: "shell",
  preserve_order: true,
  path: "provision/server/dhcp.sh"
end
```

Рис. 42: Фрагмент Vagrantfile с `server dhcp`

После этого виртуальные машины можно выключить.

```
vagrantfile validated successfully.  
PS D:\work\mavinogradova\vagrant> vagrant halt client  
==> client: Attempting graceful shutdown of VM...  
PS D:\work\mavinogradova\vagrant> vagrant halt server  
==> server: Attempting graceful shutdown of VM...  
==> server: Forcing shutdown of VM...  
PS D:\work\mavinogradova\vagrant> vagrant status  
Current machine states:  
  
server           poweroff (virtualbox)  
client           poweroff (virtualbox)  
  
This environment represents multiple VMs. The VMs are all listed  
above with their current state. For more information about a specific  
VM, run `vagrant status NAME`.  
PS D:\work\mavinogradova\vagrant> |
```

Рис. 43: Выключение виртуальных машин

Выводы

Приобретены практические навыки по установке и конфигурированию DHCP-сервера.

На виртуальной машине server:

- установлен DHCP-сервер Kea;
- настроена выдача адресов из пула `192.168.1.30–199` для виртуальной внутренней сети;
- выполнена привязка службы к интерфейсу `eth1`;
- виртуальная машина client получила IP-адрес `192.168.1.30` по DHCP, что подтверждено утилитами `ifconfig` и файлом аренд `/var/lib/kea/kea-leases4.csv`;
- настроено обновление DNS-зон при появлении новых узлов: сгенерирован TSIG-ключ `DHCP_UPDATE`, разрешено обновление зон в конфигурации `named`, сформирован файл `tsig-keys.json`, сконфигурирован сервис `kea-dhcp-ddns` и включён параметр `enable-updates` в `kea-dhcp4.conf`;